

Algebraic Exponential Integrals: Transcendence and Linear Relations

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Abstract

Let $q \in \overline{\mathbb{Q}}[x]$ be nonconstant, and let $\Delta = \sum_{\alpha} d_{\alpha}[\alpha]$ be a finite degree-zero zero-cycle on $\mathbb{A}^1(\overline{\mathbb{Q}})$ with coefficients in $\overline{\mathbb{Q}}$. If Γ is any finite piecewise C^1 one-chain in $\mathbb{C}$ with $\partial\Gamma = \Delta$, put

$$I_{q,\Delta}(z) = \int_{\Gamma} e^{zq(x)} dx.$$

We prove that every nonzero function $I_{q,\Delta}$ is a purely transcendental E -function: for every $\xi \in \overline{\mathbb{Q}}^{\times}$, $I_{q,\Delta}(\xi)$ is transcendental. Equivalently,

$$I_{q,\Delta}(1) \in \overline{\mathbb{Q}} \iff I_{q,\Delta} \equiv 0.$$

In particular, $\int_A^B e^{q(x)} dx$ is transcendental for every pair of distinct algebraic endpoints. For a fixed polynomial q , we also determine all $\overline{\mathbb{Q}}$ -linear relations among finite families of these values: they are exactly the specializations of constant-coefficient functional relations, and the latter are characterized by vanishing of explicit inverse-branch densities on a finite tree in the target of q .

The proof combines Beukers' lifting theorem with a homogeneous moment connection. A Fourier–Laplace description proves semisimplicity, while an explicit residue calculation shows that every rational horizontal endomorphism is constant. An algebraic value therefore forces an exponential-polynomial identity. Constant-coefficient annihilation and a Cauchy-transform argument then force the original exponential period function to vanish identically.

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1 Introduction and main results

Following Fresán and Jossen, we use *exponential period function* for a parameter-dependent integral of the form $\int_{\gamma} e^{-zf} \omega$ with algebraic data [9]. The present functions form a one-dimensional relative family: $X = \mathbb{A}^1$, the regular function is $-q$, the algebraic differential form is dx , and the integration chain has finite boundary. This is compatible with the general terminology of exponential periods [5]. We refer simply to the polynomial q and to its values and critical values.

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Definition 1.1 (Zero-cycles and exponential period functions). A finite zero-cycle on $\mathbb{A}^1(\overline{\mathbb{Q}})$ with coefficients in $\overline{\mathbb{Q}}$ is a formal sum

$$\Delta = \sum_{\alpha \in S} d_\alpha [\alpha], \quad S \subset \overline{\mathbb{Q}} \text{ finite}, \quad d_\alpha \in \overline{\mathbb{Q}}.$$

Its degree is $\deg \Delta = \sum_{\alpha \in S} d_\alpha$. Suppose $\deg \Delta = 0$. Choose a finite piecewise C^1 one-chain Γ in $\mathbb{C}$, with complex coefficients, such that $\partial \Gamma = \Delta$, and set

$$I_{q,\Delta}(z) = \int_{\Gamma} e^{zq(x)} dx. \quad (1.1)$$

For every z , the one-form $e^{zq(x)} dx$ has an entire primitive in x . Thus (1.1) is independent of the chosen Γ , and compactness of Γ shows that $I_{q,\Delta}$ is entire in z .

Theorem 1.2 (Algebraic-value rigidity). *Let $q \in \overline{\mathbb{Q}}[x]$ be nonconstant, and let Δ be a finite degree-zero zero-cycle on $\mathbb{A}^1(\overline{\mathbb{Q}})$ with coefficients in $\overline{\mathbb{Q}}$. Then*

$$I_{q,\Delta}(1) \in \overline{\mathbb{Q}} \iff I_{q,\Delta}(z) \equiv 0. \quad (1.2)$$

In the equivalent cases, $I_{q,\Delta}(1) = 0$.

The fixed evaluation point 1 entails no loss: a nonzero algebraic parameter is already part of the polynomial in the exponent.

Corollary 1.3 (Pure transcendence and algebraic zeros). *If $I_{q,\Delta}$ is not identically zero, then*

$$I_{q,\Delta}(\xi) \notin \overline{\mathbb{Q}} \quad (\xi \in \overline{\mathbb{Q}}^\times).$$

In particular, $I_{q,\Delta}$ has no nonzero algebraic zero. As an E -function, it is purely transcendental in the standard sense.

Proof. For $\xi \in \overline{\mathbb{Q}}^\times$,

$$I_{\xi q,\Delta}(z) = I_{q,\Delta}(\xi z).$$

The right side is nonzero as an entire function if and only if $I_{q,\Delta}$ is nonzero. Apply Theorem 1.2 to the polynomial ξq . $\square$

Corollary 1.4 (A single algebraic exponential integral). *Let $q \in \overline{\mathbb{Q}}[x]$ be nonconstant and let $A, B \in \overline{\mathbb{Q}}$ with $A \neq B$. Then, for every $\xi \in \overline{\mathbb{Q}}^\times$,*

$$\int_A^B e^{\xi q(x)} dx$$

is transcendental and nonzero.

Proof. For $\Delta = [B] - [A]$, one has $I_{q,\Delta}(0) = B - A \neq 0$, so $I_{q,\Delta}$ is not identically zero. Apply Corollary 1.3. $\square$

For a fixed E -function, Adamczewski and Rivoal gave an algorithm that computes all nonzero algebraic arguments at which its value is algebraic [1]; Bostan, Rivoal, and Salvy made the required minimal-equation computations effective in practice [4]. The point of Theorem 1.2 is different: it gives a uniform answer for the entire family indexed by (q, Δ) and identifies exactly when the

function itself vanishes. Delaygue's Lindemann–Weierstrass theorem gives linear independence under a separation condition on the finite singularity sets of the associated G -series [7]. The functions attached to a fixed q arise from one common moment connection and can have overlapping singularity data; the results below require no such separation and describe the whole relation space.

There is classical overlap in special cases. For example,

$$\int_0^A e^{cx^n} dx = A {}_1F_1\left(\frac{1}{n}; 1 + \frac{1}{n}; cA^n\right),$$

so monomial exponents belong to the hypergeometric E -function setting; see, for example, Delaygue's discussion of hypergeometric values [7]. Our argument treats arbitrary polynomials, arbitrary algebraic degree-zero zero-cycles, and all functional relations among a fixed- q family. The no-algebraic-zero consequence also connects with recent work of Fischler and Rivoal on zeros of E -functions [8].

The proof has three parts. First, moments of (1.1) satisfy one explicit inhomogeneous E -function system. Second, the homogeneous connection is semisimple and every rational horizontal endomorphism is constant; this converts a polynomial-coefficient relation supplied by Beukers' theorem into a constant-coordinate relation. Third, an exponential-polynomial identity produces vanishing polynomial moments, and a Cauchy transform on a finite tree forces all inverse-branch densities to vanish.

Scope of the differential form. The fixed form dx is essential. For $q(x) = x(x-1)$ and $r(x) = 2x^2 - x + 1$, one has

$$r(x)e^{q(x)} = \frac{d}{dx}(xe^{q(x)}), \quad \int_0^1 r(x)e^{q(x)} dx = 1.$$

Yet the function $\int_0^1 r(x)e^{zq(x)} dx$ is not identically zero: its value at $z = 0$ is $7/6$. Thus Theorem 1.2 does not extend unchanged to arbitrary polynomial amplitudes.

2 The moment system

We use the standard definition: an entire function $F(z) = \sum_{m \geq 0} a_m z^m / m!$ is an E -function if its coefficients are algebraic, $h(a_0, \dots, a_m) = O(m)$, and F satisfies a nonzero linear differential equation over $\overline{\mathbb{Q}}[z]$.

Assume throughout this section that $n = \deg q \geq 2$, and put

$$V = \mathbb{C}[x]_{< n-1}.$$

For $0 \leq j \leq n-2$, divide

$$q(x)x^j = h_j(x)q'(x) + r_j(x), \quad \deg r_j < n-1. \quad (2.1)$$

Define endomorphisms C, D of V by

$$C(x^j) = r_j, \quad D(x^j) = -h'_j.$$

Let $C = C^\top$ and $D = D^\top$ in the monomial basis $1, x, \dots, x^{n-2}$.

Lemma 2.1 (Residue spectrum). *The matrix D is triangular, with*

$$\text{Spec}(D) = \left\{ -\frac{1}{n}, -\frac{2}{n}, \dots, -\frac{n-1}{n} \right\}. \quad (2.2)$$

Proof. Comparison of the leading terms in (2.1) gives

$$h_j(x) = \frac{x^{j+1}}{n} + O(x^j).$$

Consequently, $-h'_j(x) = -(j+1)x^j/n + O(x^{j-1})$. $\square$

Fix a chain Γ with boundary Δ , and put

$$U_j(z) = \int_{\Gamma} x^j e^{zq(x)} dx, \quad \mathbf{U} = (U_0, \dots, U_{n-2})^{\top}. \quad (2.3)$$

Let

$$\Lambda = \{0\} \cup q(\text{supp } \Delta), \quad (2.4)$$

with repetitions removed, and define

$$\mathbf{h}(x) = (h_0(x), \dots, h_{n-2}(x))^{\top}, \quad \mathbf{b}_{\lambda} = \sum_{\substack{\alpha \in \text{supp } \Delta \\ q(\alpha) = \lambda}} d_{\alpha} \mathbf{h}(\alpha) \quad (\lambda \in \Lambda).$$

The empty sum is zero; in particular, $\mathbf{b}_0 = 0$ when no endpoint has q -value zero.

Proposition 2.2 (Moment system). *The moment vector satisfies*

$$\mathbf{U}' = \left(C + \frac{D}{z} \right) \mathbf{U} + \frac{1}{z} \sum_{\lambda \in \Lambda} \mathbf{b}_{\lambda} e^{\lambda z}. \quad (2.5)$$

Each U_j is an E -function.

Proof. Integration by parts in (2.1) gives, for $z \neq 0$,

$$U'_j = \int_{\Gamma} \left(r_j - \frac{h'_j}{z} \right) e^{zq} dx + \frac{1}{z} \sum_{\alpha \in \text{supp } \Delta} d_{\alpha} h_j(\alpha) e^{zq(\alpha)},$$

which is (2.5). Both sides are meromorphic at 0, and the identity holds there by continuation.

Write

$$U_j(z) = \sum_{m \geq 0} u_{j,m} \frac{z^m}{m!}, \quad u_{j,m} = \int_{\Gamma} x^j q(x)^m dx,$$

and

$$q(x)^m = \sum_{k=0}^{nm} \gamma_{m,k} x^k.$$

A polynomial primitive gives the exact formula

$$u_{j,m} = \sum_{k=0}^{nm} \gamma_{m,k} \frac{\sum_{\alpha \in \text{supp } \Delta} d_{\alpha} \alpha^{j+k+1}}{j+k+1}. \quad (2.6)$$

Thus every $u_{j,m}$ is algebraic.

We record the arithmetic growth explicitly. Let K be a number field containing the coefficients of q and all α, d_α . Uniformly over embeddings $\sigma : K \hookrightarrow \mathbb{C}$, integration of a primitive along the segment from 0 to $\sigma(\alpha)$ gives

$$|\sigma(u_{j,m})| \leq A^{m+j+2} \quad (2.7)$$

for a constant $A \geq 1$. Choose an integer $d \geq 1$ such that d times every coefficient of q , every α , and every d_α is an algebraic integer. For $m \geq 0$, set

$$\delta_m = d^{(n+1)m+j+2} \text{lcm}(1, 2, \dots, nm + j + 1). \quad (2.8)$$

Formula (2.6) shows that $\delta_m u_{j,\ell}$ is an algebraic integer for every $0 \leq \ell \leq m$: indeed, $d^\ell \gamma_{\ell,k}$, dd_α , and $d^{j+k+1} \alpha^{j+k+1}$ are integral, while $k \leq n\ell \leq nm$, and the total exponent of d needed for each term satisfies $\ell + j + k + 2 \leq (n+1)m + j + 2$. Since $\log \text{lcm}(1, \dots, N) = O(N)$, (2.7) and (2.8) imply

$$h(u_{j,0}, \dots, u_{j,m}) = O(m).$$

Finally, (2.5) shows that the $\overline{\mathbb{Q}}(z)$ -span of the U_j and the finitely many functions $e^{\lambda z}$ is stable under differentiation. Hence each U_j satisfies a nonzero scalar differential equation over $\overline{\mathbb{Q}}[z]$. This proves all defining properties of an E -function. $\square$

When $q(x) = ax + b$ is linear, the explicit identity

$$I_{q,\Delta}(z) = \frac{1}{az} \sum_{\alpha \in \text{supp } \Delta} d_\alpha e^{zq(\alpha)} \quad (z \neq 0) \quad (2.9)$$

shows directly that $I_{q,\Delta}$ is also an E -function.

We use the following linear form of Beukers' refinement of the Siegel–Shidlovskii theorem [3, Theorem 3.2 and Lemma 3.1]. No linear independence hypothesis on the coordinates is required.

Theorem 2.3 (Beukers, linear lifting). *Let $\mathbf{F} = (F_1, \dots, F_N)^\top$ be a vector of E -functions satisfying $\mathbf{F}' = A(z)\mathbf{F}$ with $A(z) \in M_N(\overline{\mathbb{Q}}(z))$, and suppose that A has no pole at $z = 1$. For every relation*

$$a_1 F_1(1) + \dots + a_N F_N(1) = 0, \quad a_i \in \overline{\mathbb{Q}},$$

there are polynomials $p_i \in \overline{\mathbb{Q}}[z]$ such that

$$\sum_{i=1}^N p_i(z) F_i(z) = 0, \quad p_i(1) = a_i \quad (1 \leq i \leq N).$$

3 The homogeneous moment connection

The homogeneous part of (2.5) is

$$\mathbf{Y}' = \left(C + \frac{D}{z} \right) \mathbf{Y}. \quad (3.1)$$

Proposition 3.1 (Semisimplicity). *The differential module over $\mathbb{C}(z)$ defined by (3.1) is semisimple.*

Proof. Let Σ be the finite set of critical values of q , and put $V_0 = \mathbb{A}_t^1 \setminus \Sigma$. Over V_0 , the finite map $q : q^{-1}(V_0) \rightarrow V_0$ defines a rank- n permutation local system $\mathcal{L}$. Its monodromy group is finite, so $\mathcal{L}$ is semisimple by Maschke's theorem. The trace map splits it as

$$\mathcal{L} \simeq \mathbb{C}_{V_0} \oplus \mathcal{L}^0, \quad \mathcal{L}^0 = \ker(\text{Tr} : \mathcal{L} \rightarrow \mathbb{C}_{V_0}).$$

The map q is finite, hence proper and small. The small-map intermediate-extension identity [6, Remark 4.2.4] gives

$$Rq_* \mathbb{C}_{\mathbb{A}_x^1}[1] \simeq \text{IC}(\mathcal{L}). \quad (3.2)$$

Intermediate extension preserves direct sums and takes irreducible local systems to simple perverse sheaves. Thus the semisimplicity of $\mathcal{L}$ makes this perverse sheaf semisimple. Under the Riemann–Hilbert correspondence and its compatibility with proper direct image, the regular holonomic $\mathcal{D}$ -module $q_+ \mathcal{O}_{\mathbb{A}_x^1}$ is therefore semisimple; the constant summand of $\mathcal{L}$ corresponds to $\mathcal{O}_{\mathbb{A}_t^1}$ [11].

Use a Fourier variable τ and the Weyl-algebra convention

$$t \longmapsto -\partial_\tau, \quad \partial_t \longmapsto \tau. \quad (3.3)$$

This is the inverse convention to the transform with kernel $e^{t\tau}$ in [14, §1.3.b]; our convention corresponds to the kernel $e^{-t\tau}$. Fourier transformation is an exact equivalence on holonomic Weyl modules and preserves simple objects and direct sums. The transform of $\mathcal{O}_{\mathbb{A}_t^1}$ is supported at $\tau = 0$, hence disappears after localization to $\mathbb{C}(\tau)$. A nonzero generic localization of a simple Weyl module remains irreducible. Indeed, its $\mathbb{C}[\tau] \setminus \{0\}$ -torsion is a Weyl submodule: if $p(\tau)m = 0$, then $p(\tau)^2 \partial_\tau m = 0$. Thus a simple module whose localization is nonzero embeds in that localization. Any nonzero $\mathbb{C}(\tau)$ -differential submodule meets the original module after clearing a denominator, and simplicity then forces equality. It follows that the generic Fourier transform of $q_+ \mathcal{O}_{\mathbb{A}_x^1}$ is semisimple.

For completeness, we identify this generic transform explicitly. Factor q as its graph embedding followed by projection to the t -line. A graph generator δ_q satisfies

$$(t - q(x))\delta_q = 0, \quad (\partial_x + q'(x)\partial_t)\delta_q = 0.$$

After (3.3), the transformed generator satisfies

$$(\partial_\tau + q(x))\widehat{\delta}_q = 0, \quad (\partial_x + \tau q'(x))\widehat{\delta}_q = 0.$$

In particular, $\partial_x(p\widehat{\delta}_q) = (p' - \tau q'p)\widehat{\delta}_q$: the relation on the generator induces a minus sign in the differential on coefficient polynomials. The generic direct image along the x -line is therefore the degree-zero cohomology of

$$\mathbb{C}(\tau)[x] \xrightarrow{\partial_x - \tau q'(x)} \mathbb{C}(\tau)[x],$$

with τ -connection induced by $\partial_\tau - q(x)$. The displayed map is injective: for nonzero p , the term $-\tau q'p$ has larger x -degree than p' . Hence the generic transform is

$$\mathbb{C}(\tau)[x]/(\partial_x - \tau q'(x))\mathbb{C}(\tau)[x], \quad \nabla_{\partial_\tau}[p] = [\partial_\tau p - qp]. \quad (3.4)$$

This is the algebraic Fourier–Laplace direct-image construction of [14, §1.3.b], with the inverse convention fixed in (3.3).

Pulling (3.4) back by $\tau = -z$ gives

$$\mathcal{M}_q = \mathbb{C}(z)[x]/(\partial_x + zq'(x))\mathbb{C}(z)[x], \quad \nabla_{\partial_z}[p] = [\partial_z p + qp]. \quad (3.5)$$

The connection descends to this quotient because

$$[\partial_z + q, \partial_x + zq'] = q' - q' = 0. \quad (3.6)$$

The classes of $1, x, \dots, x^{n-2}$ form a basis: the relation $[zq'p] = -[p']$ reduces every higher monomial, while degree comparison proves independence. In this basis, (2.1) gives the connection matrix $C + D/z$. The period system (3.1) is the corresponding dual system, and duality preserves semisimplicity. $\square$

Proposition 3.2 (Rational endomorphism rigidity). *If $P(z) \in M_{n-1}(\mathbb{C}(z))$ satisfies*

$$P' = \left[C + \frac{D}{z}, P \right], \quad (3.7)$$

then P is constant. The horizontal endomorphism algebra of (3.1) is therefore

$$Z(C) \cap Z(D).$$

The same constancy conclusion holds for the dual module.

Proof. A pole at $z_0 \neq 0$ is impossible: differentiation raises its order by one, whereas the right side of (3.7) does not. At 0, a leading term $z^{-m}P_{-m}$ with $m \geq 1$ would give

$$-mP_{-m} = [D, P_{-m}].$$

By (2.2), every eigenvalue of $\text{ad } D$ lies strictly between -1 and 1 , so the negative integer $-m$ cannot occur. Thus P has no finite pole and is polynomial.

It remains to exclude positive degree. Transpose (3.7) and write $Q = P^\top$. The operator C is multiplication by q on $\mathbb{C}[x]/(q')$. For a critical point τ , put

$$m_\tau = \text{ord}_\tau q', \quad \lambda_\tau = q(\tau),$$

and define

$$g_{\tau,k}(x) = \frac{q'(x)}{(x - \tau)^k}, \quad 1 \leq k \leq m_\tau.$$

The Chinese remainder theorem shows that all the $g_{\tau,k}$ form a basis of V . Moreover,

$$Cg_{\tau,k} = \lambda_\tau g_{\tau,k},$$

because

$$qg_{\tau,k} = \frac{q - \lambda_\tau}{(x - \tau)^k} q' + \lambda_\tau g_{\tau,k}.$$

Thus C is semisimple, with eigenspaces indexed by critical values.

Let E_λ be the λ -eigenspace of C , let π_λ be its spectral projection, and put

$$\overline{D}_\lambda = \pi_\lambda D|_{E_\lambda}.$$

Fix τ with $\lambda_\tau = \lambda$, and write $m = m_\tau$. Define

$$F_{\tau,0} = 0, \quad F_{\tau,k} = \text{span}_{\mathbb{C}}\{g_{\tau,1}, \dots, g_{\tau,k}\} \quad (1 \leq k \leq m).$$

We claim the explicit flag relation

$$\overline{D}_\lambda g_{\tau,k} + \frac{m+1-k}{m+1} g_{\tau,k} \in F_{\tau,k-1} \quad (1 \leq k \leq m). \quad (3.8)$$

To prove it, write $y = x - \tau$ and $q - \lambda = ay^{m+1} + O(y^{m+2})$, with $a \neq 0$. Euclidean division gives

$$Dg_{\tau,k} = - \left(\frac{q - \lambda}{(x - \tau)^k} \right)',$$

so

$$\begin{aligned} g_{\tau,k} &= (m+1)ay^{m-k} + O(y^{m-k+1}), \\ Dg_{\tau,k} &= -(m+1-k)ay^{m-k} + O(y^{m-k+1}). \end{aligned}$$

Their combination in (3.8) therefore vanishes at τ to order at least $m - k + 1$. If $\sigma \neq \tau$ has the same critical value, then $(x - \tau)^{-k}$ is regular at σ , while $q - \lambda$ vanishes there to order $m_\sigma + 1$. Hence $Dg_{\tau,k}$ vanishes at σ to order at least m_σ , and its component in that factor of $\mathbb{C}[x]/(q')$ is zero.

Under the Chinese remainder decomposition, $F_{\tau,k-1}$ consists precisely of the classes supported in the τ -factor and divisible there by y^{m-k+1} . Projection by π_λ removes all factors over other critical values. The preceding vanishing statements thus prove (3.8). In particular, every $F_{\tau,k}$ is $\overline{D}_\lambda$ -invariant. On each $F_{\tau,m}$ the matrix in the ordered basis $g_{\tau,1}, \dots, g_{\tau,m}$ is triangular, with diagonal

$$-\frac{m}{m+1}, -\frac{m-1}{m+1}, \dots, -\frac{1}{m+1}. \quad (3.9)$$

As τ ranges over the critical points with value λ , these lists give the full spectrum of $\overline{D}_\lambda$, with multiplicities. In particular, all its eigenvalues lie in $(-1, 0)$, including when distinct critical points have the same value and different ramification indices.

Suppose

$$Q(z) = Q_d z^d + Q_{d-1} z^{d-1} + \dots, \quad d \geq 1, \quad Q_d \neq 0.$$

The top two coefficients of the transposed equation give

$$[Q_d, C] = 0, \quad dQ_d = [Q_{d-1}, C] + [Q_d, D]. \quad (3.10)$$

The first identity makes Q_d block diagonal for the eigenspace decomposition of C . Choose a critical value λ for which $Q_{d,\lambda} \neq 0$; the value $\lambda = 0$ is allowed. Projecting the second identity to this block yields

$$dQ_{d,\lambda} = [Q_{d,\lambda}, \overline{D}_\lambda].$$

After triangularizing $\overline{D}_\lambda$, the eigenvalues of the commutator operator on the right are pairwise differences of the numbers in (3.9). They all lie strictly between -1 and 1 , so none equals the positive integer d . This contradiction proves that $d = 0$.

A constant solution of (3.7) commutes with both C and D , and the converse is immediate. The calculation for a right-acting endomorphism of the dual gives the same equation. $\square$

Corollary 3.3 (Constant splitting). *Every differential submodule of the dual of (3.1) has the form*

$$W \otimes_{\mathbb{C}} \mathbb{C}(z)$$

for a constant subspace $W \subseteq \mathbb{C}^{1 \times (n-1)}$ invariant under right multiplication by C and D .

Proof. By Proposition 3.1, a differential submodule has a differential complement and hence a horizontal idempotent projector. By Proposition 3.2, this projector is constant. Its image is the required subspace W ; substituting the constant projector into (3.7) shows that W is C - and D -invariant. $\square$

4 From an algebraic value to an exponential polynomial

Put

$$\mathcal{E} = \text{span}_{\mathbb{C}(z)}\{e^{\lambda z} : \lambda \in \Lambda\}.$$

Proposition 4.1. *If $U_0(1) \in \overline{\mathbb{Q}}$, then*

$$U_0(z) = \sum_{\lambda \in \Lambda} p_\lambda(z) e^{\lambda z} \quad (4.1)$$

with $p_\lambda \in \overline{\mathbb{Q}}[z]$.

Proof. Adjoin the exponential coordinates $e^{\lambda z}$ to the moment vector. By (2.5), the resulting vector of E -functions satisfies a homogeneous block system whose only finite pole is at $z = 0$. Let $a = U_0(1) \in \overline{\mathbb{Q}}$. Applying Theorem 2.3 to $U_0(1) - a \cdot 1 = 0$ gives a polynomial row $\phi(z)$ and polynomials $c_\lambda(z)$ such that

$$\phi(z)\mathbf{U}(z) + \sum_{\lambda \in \Lambda} c_\lambda(z) e^{\lambda z} = 0, \quad \phi(1) = (1, 0, \dots, 0). \quad (4.2)$$

Define

$$\mathcal{R} = \{\psi \in \mathbb{C}(z)^{1 \times (n-1)} : \psi \mathbf{U} \in \mathcal{E}\}.$$

If $A(z) = C + D/z$ and $\mathbf{f} = z^{-1} \sum_\lambda \mathbf{b}_\lambda e^{\lambda z}$, then

$$(\psi \mathbf{U})' = (\psi' + \psi A) \mathbf{U} + \psi \mathbf{f}.$$

Since $\mathcal{E}$ is stable under differentiation, this proves that $\mathcal{R}$ is a differential submodule of the dual homogeneous connection. By Corollary 3.3, $\mathcal{R} = W \otimes_{\mathbb{C}} \mathbb{C}(z)$ for a constant C - and D -invariant subspace W . Equation (4.2) puts ϕ in $\mathcal{R}$. Every constant linear form annihilating W therefore annihilates $\phi(z)$ identically, and hence also annihilates $\phi(1)$. Thus $\phi(1) \in W$, without specializing any rational basis. Hence the constant row $\mathbf{e}_0 = (1, 0, \dots, 0)$ belongs to W , and $U_0 = \mathbf{e}_0 \mathbf{U}$ belongs to $\mathcal{E}$.

Choose a full-row-rank constant matrix S whose rows form a basis of W , and put $\mathbf{V} = S \mathbf{U}$. There are constant matrices C_W, D_W such that

$$SC = C_W S, \quad SD = D_W S.$$

Every component of $\mathbf{V}$ belongs to $\mathcal{E}$. The functions $e^{\lambda z}$ are linearly independent over $\mathbb{C}(z)$, so there is a unique expansion

$$\mathbf{V} = \sum_{\lambda \in \Lambda} \mathbf{r}_\lambda(z) e^{\lambda z}, \quad \mathbf{r}_\lambda \in \mathbb{C}(z)^{\dim W}.$$

Equating exponential coefficients in the restricted moment system gives

$$\mathbf{r}'_\lambda = \left(C_W - \lambda I + \frac{D_W}{z} \right) \mathbf{r}_\lambda + \frac{S \mathbf{b}_\lambda}{z}. \quad (4.3)$$

A pole away from 0 is impossible by order comparison. If $\mathbf{r}_\lambda = z^{-m}v + O(z^{-m+1})$ at 0, with $m \geq 1$, then (4.3) gives

$$-mv = D_W v.$$

The eigenvalues of the restriction D_W are among those of D , hence lie strictly between -1 and 0 by (2.2); none is a negative integer. Thus every $\mathbf{r}_\lambda$ is polynomial, and so are the coefficients p_λ in (4.1).

It remains to show that their coefficients are algebraic. Choose an integer $d \geq 0$ bounding the degrees of all the p_λ , write $p_\lambda(z) = \sum_{k=0}^d a_{\lambda,k} z^k$, and put $M = (d+1)|\Lambda|$. Differentiating (4.1) at 0 gives

$$U_0^{(m)}(0) = \sum_{\lambda \in \Lambda} \sum_{k=0}^d a_{\lambda,k} \left. \frac{d^k}{dt^k} t^m \right|_{t=\lambda} \quad (0 \leq m < M). \quad (4.4)$$

The square coefficient matrix has algebraic entries and is invertible. Indeed, its transpose is the Hermite evaluation map

$$\overline{\mathbb{Q}}[t]_{<M} \longrightarrow \overline{\mathbb{Q}}^M, \quad f \longmapsto (f^{(k)}(\lambda))_{\lambda \in \Lambda, 0 \leq k \leq d}.$$

A polynomial in its kernel is divisible by $\prod_{\lambda \in \Lambda} (t - \lambda)^{d+1}$, of degree M , so must vanish. Thus the map is injective and hence invertible. All derivatives $U_0^{(m)}(0)$ are algebraic by (2.6); solving (4.4) yields $a_{\lambda,k} \in \overline{\mathbb{Q}}$. Therefore $p_\lambda \in \overline{\mathbb{Q}}[z]$. $\square$

5 Moments, Cauchy transforms, and functional relations

We first isolate the geometric representation used by the analytic argument.

Lemma 5.1 (Tree representation). *Let $q \in \mathbb{C}[x]$ be nonconstant and let Δ be a finite degree-zero zero-cycle on $\mathbb{C}$ with complex coefficients. There exist a finite embedded tree T in the t -plane, containing all finite critical values of q and all values $q(\alpha)$ for $\alpha \in \text{supp } \Delta$, and a one-chain $\tilde{\Gamma}$ in the graph $q^{-1}(T)$ such that $\partial \tilde{\Gamma} = \Delta$.*

Orient every open edge e of T . Over e , write $\tilde{\Gamma}$ as a linear combination of analytic inverse branches $x_{e,i}$ of q , with coefficients $f_{e,i}$, and set

$$\psi_e(t) = \sum_i f_{e,i} x'_{e,i}(t) = \sum_i \frac{f_{e,i}}{q'(x_{e,i}(t))}. \quad (5.1)$$

Then ψ_e is analytic on the open edge, has at worst integrable singularities at its endpoints, and for every entire function F ,

$$\int_{\Gamma} F(q(x)) dx = \sum_e \int_e F(t) \psi_e(t) dt, \quad (5.2)$$

where Γ is any one-chain with boundary Δ .

Proof. Choose a regular value c of q , distinct from all endpoint values, and join c to the distinct finite critical values and endpoint values by simple analytic arcs with pairwise disjoint interiors. Their union is a finite tree T . The inverse image of the critical-value arms is the standard polynomial constellation. It is connected because the monodromy of the polynomial covering on a regular fiber is transitive [12, §2.1]. The remaining arms attach to this connected subgraph, so $q^{-1}(T)$ is connected.

Subdivide the graph so that every point of $\text{supp } \Delta$ is a vertex. Fix a vertex x_* and choose an edge path P_α from x_* to each $\alpha \in \text{supp } \Delta$. Then

$$\tilde{\Gamma} = \sum_{\alpha \in \text{supp } \Delta} d_\alpha P_\alpha$$

has boundary Δ , because $\deg \Delta = 0$. The inverse-branch description on open edges gives (5.1). Near a ramified endpoint of index r , one has $x'_{e,i}(t) = O(|t - t_0|^{1/r-1})$, which is integrable.

The one-form $F(q(x))dx$ is entire in x , so its integral depends only on the boundary. Replacing Γ by $\tilde{\Gamma}$ and changing variables $t = q(x)$ edge by edge gives (5.2). $\square$

Theorem 5.2 (Moment rigidity and the edge-density criterion). *In the setting of Lemma 5.1, let $0 \neq H \in \mathbb{C}[t]$. If*

$$\int_{\Gamma} q(x)^m H(q(x)) dx = 0 \quad (m \geq 0), \quad (5.3)$$

then every density ψ_e in (5.1) vanishes identically. Consequently,

$$I_{q,\Delta}(z) \equiv 0.$$

For any fixed tree representation,

$$I_{q,\Delta} \equiv 0 \iff \psi_e \equiv 0 \text{ for every open edge } e \subset T. \quad (5.4)$$

Proof. Every differential $q(x)^m H(q(x))dx$ is polynomial and has a polynomial primitive, so (5.3) remains true after replacing Γ by $\tilde{\Gamma}$. For $s \in \mathbb{C} \setminus T$, define

$$\mathcal{C}_H(s) = \int_{\tilde{\Gamma}} \frac{H(q(x))}{s - q(x)} dx.$$

For sufficiently large $|s|$, the geometric series for $(s - q(x))^{-1}$ converges uniformly on the compact chain, and (5.3) gives $\mathcal{C}_H(s) = 0$. The complement of a finite embedded tree is connected, so the identity theorem yields

$$\mathcal{C}_H \equiv 0 \quad \text{on } \mathbb{C} \setminus T. \quad (5.5)$$

By (5.2),

$$\mathcal{C}_H(s) = \sum_e \int_e \frac{H(t)\psi_e(t)}{s - t} dt. \quad (5.6)$$

Fix an interior point t_0 of an edge e with $H(t_0) \neq 0$, and choose a compact analytic subarc $\eta \Subset e$ containing t_0 . Contributions to (5.6) from outside η extend holomorphically across a small disk about t_0 . Put $\rho(t) = H(t)\psi_e(t)$ and

$$J(s) = \int_{\eta} \frac{\rho(t)}{s - t} dt.$$

If a, b are the oriented endpoints of η , then locally

$$J(s) = \rho(s)(\log(s - a) - \log(s - b)) - \int_{\eta} \frac{\rho(t) - \rho(s)}{t - s} dt.$$

The second term is holomorphic across η , while the two lateral branches of the logarithmic term differ by $2\pi i \varepsilon_e \rho(t_0)$, where $\varepsilon_e \in \{\pm 1\}$. By (5.5), both lateral values of the full Cauchy transform

are zero. Hence $H(t_0)\psi_e(t_0) = 0$, and therefore $\psi_e(t_0) = 0$. The point t_0 was arbitrary outside the finite zero set of H ; analyticity gives $\psi_e \equiv 0$ on every open edge.

Taking $F(t) = e^{zt}$ in (5.2) now gives $I_{q,\Delta} \equiv 0$. Conversely, if $I_{q,\Delta} \equiv 0$, its Taylor coefficients give (5.3) with $H = 1$, so the preceding argument forces all densities to vanish. The reverse implication in (5.4) follows directly from (5.2). $\square$

This argument is a local Cauchy-transform form of techniques used in the polynomial moment problem and for Cauchy-type integrals of algebraic functions [12, 13, 10].

6 Proofs and linear relations

Proof of Theorem 1.2. The reverse implication is immediate. Assume that $I_{q,\Delta}(1) = \gamma \in \overline{\mathbb{Q}}$.

Suppose first that $q(x) = ax + b$ is linear. Formula (2.9) at $z = 1$ gives

$$\sum_{\alpha \in \text{supp } \Delta} d_\alpha e^{q(\alpha)} - a\gamma e^0 = 0. \quad (6.1)$$

The values $q(\alpha)$ are pairwise distinct. Combine the two coefficients of e^0 if one endpoint satisfies $q(\alpha) = 0$. The linear form of the Lindemann–Weierstrass theorem [2, Chapter 1] then makes every coefficient in (6.1) zero. Thus $d_\alpha = 0$ for all endpoints with $q(\alpha) \neq 0$. If $q(\alpha_0) = 0$, then $d_{\alpha_0} = a\gamma$, while $\deg \Delta = 0$ gives $d_{\alpha_0} = 0$ and hence $\gamma = 0$. If no such endpoint exists, the coefficient of e^0 gives $\gamma = 0$ directly. Therefore $\Delta = 0$ and $I_{q,\Delta} \equiv 0$.

Now suppose $\deg q \geq 2$. By Proposition 4.1,

$$U_0(z) = I_{q,\Delta}(z) = \sum_{\lambda \in \Lambda} p_\lambda(z) e^{\lambda z}.$$

If $U_0 \equiv 0$, there is nothing to prove. Otherwise set

$$N = 1 + \max_{p_\lambda \neq 0} \deg p_\lambda, \quad L = \prod_{\lambda \in \Lambda} (\partial_z - \lambda)^N,$$

and

$$H(t) = \prod_{\lambda \in \Lambda} (t - \lambda)^N.$$

Then $LU_0 = 0$, and differentiation under the integral sign gives

$$0 = LU_0(z) = \int_{\Gamma} H(q(x)) e^{zq(x)} dx.$$

Comparing Taylor coefficients at $z = 0$ gives (5.3). Apply Theorem 5.2. $\square$

Corollary 6.1 (Transfer of linear relations). *Let $q \in \overline{\mathbb{Q}}[x]$ be nonconstant, let $\Delta_1, \dots, \Delta_r$ be finite degree-zero zero-cycles on $\mathbb{A}^1(\overline{\mathbb{Q}})$ with coefficients in $\overline{\mathbb{Q}}$, and put $I_j(z) = I_{q,\Delta_j}(z)$. For $a_0, \dots, a_r \in \overline{\mathbb{Q}}$,*

$$a_0 + \sum_{j=1}^r a_j I_j(1) = 0 \quad (6.2)$$

if and only if

$$a_0 = 0, \quad \sum_{j=1}^r a_j I_j(z) \equiv 0.$$

Consequently,

$$\dim_{\overline{\mathbb{Q}}} \text{span}_{\overline{\mathbb{Q}}} \{1, I_1(1), \dots, I_r(1)\} = 1 + \dim_{\overline{\mathbb{Q}}} \text{span}_{\overline{\mathbb{Q}}} \{I_1, \dots, I_r\}.$$

Proof. Put $\Delta = \sum_j a_j \Delta_j$. Relation (6.2) says $I_{q,\Delta}(1) = -a_0 \in \overline{\mathbb{Q}}$. By Theorem 1.2, $I_{q,\Delta} \equiv 0$, and then its value at 1 gives $a_0 = 0$. The converse and the dimension identity are immediate. $\square$

The same equivalence holds with the common evaluation point 1 replaced by any $\xi \in \overline{\mathbb{Q}}^\times$: replace q by ξq and then rescale the function variable.

Corollary 6.2 (Geometric description of the relation space). *In the setting of Corollary 6.1, choose one tree T containing all critical values of q and all values $q(\text{supp } \Delta_j)$. For each j , choose a graph chain representing Δ_j as in Lemma 5.1, and denote its density on an open edge e by $\psi_{j,e}$. Then*

$$a_0 + \sum_{j=1}^r a_j I_j(1) = 0$$

if and only if

$$a_0 = 0, \quad \sum_{j=1}^r a_j \psi_{j,e} \equiv 0 \quad \text{for every open edge } e \subset T.$$

Thus all $\overline{\mathbb{Q}}$ -linear relations among the values are determined by a finite family of inverse-branch identities on T .

Proof. The graph chain for $\Delta = \sum_j a_j \Delta_j$ is the same linear combination of the chosen graph chains, and its density on e is $\sum_j a_j \psi_{j,e}$. Combine Corollary 6.1 with the exact criterion (5.4). $\square$

Corollary 6.3 (Linear independence over distinct regular values). *Let $q \in \overline{\mathbb{Q}}[x]$ be nonconstant, and let $a_0, \dots, a_r \in \overline{\mathbb{Q}}$ be such that $q(a_0), \dots, q(a_r)$ are pairwise distinct regular values of q . Then, for every $\xi \in \overline{\mathbb{Q}}^\times$,*

$$1, \quad \int_{a_0}^{a_1} e^{\xi q(x)} dx, \quad \dots, \quad \int_{a_0}^{a_r} e^{\xi q(x)} dx$$

are linearly independent over $\overline{\mathbb{Q}}$.

Proof. Absorb ξ into q . It suffices to prove that the functions

$$I_j(z) = \int_{a_0}^{a_j} e^{zq(x)} dx, \quad 1 \leq j \leq r,$$

are linearly independent. A functional relation produces a degree-zero zero-cycle

$$\Delta = \sum_{j=1}^r c_j [a_j] - \left(\sum_{j=1}^r c_j \right) [a_0]$$

whose exponential period function vanishes identically. Choose the tree so that each $q(a_i)$ is the endpoint of a separate terminal arm. Over the arm ending at $q(a_i)$, regularity and distinctness imply that the only nonzero terminal boundary coefficient can occur on the lift ending at a_i . The density there is therefore

$$\frac{d_i}{q'(x_i(t))},$$

where d_i is the coefficient of $[a_i]$ in Δ . By (5.4), this density vanishes, so $d_i = 0$ for every i . Hence all $c_j = 0$. Apply Corollary 6.1. $\square$

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